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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/244,379	04/29/2021	Shenglong Gao	179648.07101/185US01	7955

159058 7590 10/30/2023

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EXAMINER	
KWIATKOWSKA, LIDIA	

ART UNIT	PAPER NUMBER
3666	

NOTIFICATION DATE	DELIVERY MODE
10/30/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/244,379	Applicant(s) Gao et al.	
	Examiner LIDIA KWIATKOWSKA	Art Unit 3666	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 14 August 2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-22 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-22 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 29 April 2021 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Status of Claims

- This Non-**Final action** is in response **to** the applicant's **amendment** of **08/14/2023**.
- Claims **1-22** are pending and have been considered as follows.

Response to Amendment

Applicant's amendments with respect to the rejection of claims under **35 USC § 103 have been fully considered** but are moot. The examiner agrees that the prior art does not explicitly teach as it is currently recited in the claim language; "departure confirmation comprises data confirming that coordinates of the passenger electronic device" and "otherwise causing the AV to either (a) navigate to an intermediate stopping location before reaching the pickup area or (b) pass through the pickup area". Therefore, the rejection has been withdrawn; However, upon further consideration a new ground(s) of **rejection is made for Claims 1-22** over **Dyer (US Patent No. US 20210053567 A1)** in view of **Leary (US Patent No. US20200410406A1**) and **Rasmusson (US Patent No. US20180136656A1).**

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

Claim(s) 1-22 are rejected under 35 U.S.C. 103 as being unpatentable over over **Dyer (US Patent No. US 20210053567 A1)** in view of **Leary (US Patent No. US20200410406A1**) and **Rasmusson (US Patent No. US20180136656A1).**

Regarding claim 1, Dyer teaches, **a method of determining a stopping location for a vehicle before picking up a passenger, the method comprising: by one or more processors, at least one of which is a component of an autonomous vehicle (AV);** (See **Dyer paragraph 0002;** "One aspect of the disclosure provides a method of maneuvering a vehicle in an autonomous driving mode; identifying, by the one or more processors, a geographic constraint for the pullover maneuver … and maneuvering, by the one or more processors, the vehicle in the autonomous driving mode based on the determination whether to attempt to find a pullover location within the region." *(Examiner notes: pull over locations the same as stopping location)*);

receiving a ride service request that has been transmitted from a ride service application on a passenger electronic device; determining a pickup area for the ride service request, along with a loading point that is within a pickup area; (See **Dyer paragraph 0042;** "...By way of example only, client computing device 420 may be a mobile phone or a device such as a wireless-enabled PDA, a tablet PC, a wearable computing device or system, or a netbook that is capable of obtaining information via the Internet or other networks. In another example, client computing device 430 may be a wearable computing system, shown as a wristwatch as shown in FIG. 3. As an example, the user may input information using a small keyboard, a keypad, microphone, using visual signals with a camera, or a touch screen.", also **See Dyer paragraph 0076;** " For instance, the server computing devices may use the model to identify a region for a given pickup or destination location for a trip (i.e. where a passenger wants to be picked up or dropped off) computing devices 110 may send a request to the server computing devices to identify a region given a current destination for the vehicle and/or pickup or drop off location for a passenger."); **determining a route from a starting location of the AV to the pickup area; navigating the AV along the route toward the pickup area;** (See **Dyer, paragraph 0027**; "Planning system 168 may be used by computing devices 110 in order to determine and follow a route generated by a routing system 166 to a location. For instance, the routing system 166 may use map information to determine a route from a current location of the vehicle to a drop off location. The planning system 168 may periodically generate trajectories, or short-term plans for, controlling the vehicle for some period of time into the future, in order to follow the route to the destination."; also **see Dyer Figure 10** block 1050).

Dyer does not explicitly teach but Leary teaches, **and if the one or more processors have received the departure confirmation, causing the AV to navigate into the pickup area and stop at the loading point;** (See **Leary, paragraph 0079**; "If the user does not end or cancel the trip, the autonomous vehicle 102 can continue and pick up the user. Once the user is picked up by the autonomous vehicle 102, the autonomous vehicle 102 can travel towards the drop-off location configured for the trip."); **otherwise causing the AV to either (a) navigate to an intermediate stopping location before reaching the pickup area or (b) pass through the pickup area;** (See **Leary, paragraph 0084;** " FIG. 5B illustrates an example notification 520 sent to the user's client device 170 to inform the user that the autonomous vehicle 102 is searching for a pull-over location. As the autonomous vehicle 102 searches (and/or before) for a pull-over location, the autonomous vehicle 102 can send (e.g., via the internal computing system 110 and/or the remote computing system 150) the notification 520 to the client device 170 for presentation to the user at the client device 170. The notification 520 can indicate that the autonomous vehicle 102 is searching for a pull-over location. In some cases, the notification 520 can also provide other information such as, for example, an indication of one or more pull-over options identified by the autonomous vehicle 102, details about one or more pull-over options, an alert (e.g., a hazard alert, a weather alert, an update alert, etc.), a request to confirm a pull-over location, and/or any other information."; also **see Leary paragraph 0082;** "In some cases, the map 502 can also depict other pull-over options 506B-N for the user. The other pull-over options 506B-N can provide alternatives that the user may select from. If the user prefers one of the other pull-over options 506B-N more than the pull-over location 506A, the user can select a particular pull-over option 506B-N from the map 502 to trigger the

autonomous vehicle 102 to drop off the user at the other, selected pull-over option; and see **Leary paragraph 0084;** "FIG. 5B illustrates an example notification 520 sent to the user's client device 170 to inform the user that the autonomous vehicle 102 is searching for a pull-over location. As the autonomous vehicle 102 searches (and/or before) for a pull-over location, the autonomous vehicle 102 can send (e.g., via the internal computing system 110 and/or the remote computing system 150) the notification 520 to the client device 170 for presentation to the user at the client device 170. The notification 520 can indicate that the autonomous vehicle 102 is searching for a pull-over location. In some cases, the notification 520 can also provide other information such as, for example, an indication of one or more pull-over options identified by the autonomous vehicle 102, details about one or more pull-over options, an alert (e.g., a hazard alert, a weather alert, an update alert, etc.), a request to confirm a pull-over location, and/or any other information." *(Examiner notes, vehicle searching for pull-over location, choosing pull-over location from multiple choices and sending the notification to the client device about chosen pull-over location indicates that the vehicle is stopping intermediate stopping location (one of the pull-over locations) or pass through the pickup area (first pull-over location that might not be available.)*).

Both Dyer and Leary are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Leary navigating AV along the route. No new functionality would arise from the combination and the

combination would improve usability of Dyer by adding the navigating AV along the route. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Dyer does not explicitly teach but Rasmusson teaches, **before the AV reaches the pickup area, determining whether the one or more processors have received a departure confirmation from the passenger electronic device;** (See **Rasmusson, paragraph 0015**; "In particular embodiments, dynamic transportation matching system 160 may facilitate transportation for one or more users 101. Dynamic transportation matching system 160 may receive any number of ride requests from any number of users 101. When dynamic transportation matching system 160 receives a ride request, it may also receive an identifier from a user device 130 associated with user 101. The identifier may identify user 101 to dynamic transportation matching system 160 for the purpose of accessing rider information about user 101 in accordance with the rider's privacy settings."); **wherein the departure confirmation comprises data confirming that coordinates of the passenger electronic device correspond to the loading point;** (See **Rasmusson, paragraph 0038;** "At step 520, the computing device determines whether the autonomous vehicle is within a threshold proximity of a current path of a current ride of the autonomous vehicle. The current path may include the ride origin or the ride destination. The ride origin may be the location where the autonomous vehicle picks up the requestor...In particular embodiments, the ride origin includes GPS coordinates of the client device 130 associated with the requestor, a request location entered into a requestor device...").

Both Dyer and Rasmusson are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer autonomous vehicle rider drop-off to destination experience with Rasmusson passenger electronic device coordinates that correspond to the loading point. No new functionality would arise from the combination and the combination would improve usability of Leary by adding the passenger electronic device coordinates that correspond to the loading point. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 2, Modified Dyer teaches the method of claim 1 as discussed above. Dyer does not teach but Leary teaches; **and using the loading point location rules to select the loading point from a set of candidates loading points (See Leary, paragraph 0051;** "At block 204, the flow 200 can include determining (e.g., via the internal computing system 110 and/or the remote computing system 150) whether the user has adjusted the drop-off location for the trip. If the user has adjusted the drop-off location, at block 206, the flow 200 can include selecting (e.g., via the internal computing system 110 and/or the remote computing system 150) a new drop-off location based on the user's adjusted drop-off location. The flow 200 can then proceed to block 208, described below.").

Both Dyer and Leary are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Leary autonomous vehicle rider drop-off to destination experience. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the c autonomous vehicle rider drop-off to

destination experience. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Dyer does not explicitly teach, but Rasmusson teaches, **wherein determining the loading point comprises: determining a passenger identifier that is associated with the passenger electronic device;** (See **Rasmusson, paragraph 0015;** " When dynamic transportation matching system 160 receives a ride request, it may also receive an identifier from a user device 130 associated with user 101. The identifier may identify user 101 to dynamic transportation matching system 160 for the purpose of accessing rider information about user 101 in accordance with the rider's privacy settings."); **accessing a passenger profile server and querying the passenger profile server for a passenger profile that corresponds to the passenger identifier; accessing the passenger profile to extract one or more loading point location rules from the passenger profile; (See Rasmusson, paragraph 0015;** "The identifier may identify user 101 to dynamic transportation matching system 160 for the purpose of accessing rider information about user 101 in accordance with the rider's privacy settings. Rider information may be stored on one or more data stores associated by dynamic transportation matching system 160. In particular embodiments, rider information may include information about a particular rider or information about riders in the aggregate. Such information may include preferred pick-up or drop-off locations for a particular rider or for a threshold number of riders, one or more autonomous-vehicle settings of the rider or riders (such as preferred speed, preferred rates of acceleration/deceleration, preferred following distance when travelling at various speeds, etc.), music settings (e.g. preferred playlists, volume, etc.), frequent destinations for a particular rider, payment information, or any other suitable information.").

Both Dyer and Rasmusson are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Rasmusson confirmation, passenger identifier and ride instructions. No new functionality would arise from the combination and the combination would improve usability of Leary by adding the confirmation, passenger identifier and ride instructions. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 3, Modified Dyer teaches, the method of claim 2 as discussed above. Dyer further teaches, **the method of claim 2, wherein the loading point location rules comprise one or more rules to prioritize one or more of the following characteristics to determine the loading point: a lane segment that is in front of an entry of a building having a location that corresponds to a location of the passenger electronic device; a lane segment that is midway between two cross streets; or a lane segment that is as close as possible to the cross street that is next in the direction of traffic while remaining a threshold distance away from that cross street; (see Dyer paragraph 0073 – 0075**; "In some instances, the qualities of the highest ranked region may be compared by the computing devices 110 with qualities of a region in which the vehicle 100 is currently driving or pulled over in order to determine whether the vehicle should move to the highest ranked region. This may be helpful, for instance, if a vehicle is currently pulled over and waiting for a passenger, and the situation changes. As an example, if another vehicle approaches who may need to nudge around the vehicle, the vehicle's computing devices may do the comparison in order to determine whether it is beneficial to the vehicle, the passenger, and/or the another vehicle to move to another region.

In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously because of the

inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation).
In some instances, real time information about the availability of pullover locations observed by a vehicle, such as any of vehicles 100, 100A, 100B, 100C, may be shared with the dispatching server computing devices and/or other vehicles of the fleet. This information may be used to update the model or in conjunction with the lists of qualities to rank the regions. This information could also be used to directly update the values for the lists of qualities or as some form of additional cost in the ranking. For example. if a region (such as a block of a street) typically has a high number of available pullover locations, but the block is full of parked vehicles, this information may be observed by one vehicle and shared with the server computing devices before another vehicle gets a request for a pick up or drop off within that region. As another example, if a vehicle blocks a driveway (that may have been considered rarely used) causes inconvenience to another road user, this information may be shared and used to avoid repeating the same behavior at the same driveway.").

Regarding claim 4, Modified Dyer teaches, the method of claim 1 as discussed above. Dyer further teaches, **further comprising, by the one or more processors, determining the intermediate stopping location as a location that: is along the route between a current location of the AV and the loading point; and satisfies each of a plurality of permissible stopping location criteria; (See Dyer paragraph 0075;** "In some instances, real time information about the availability of pullover locations observed by a vehicle, such as any of vehicles 100, 100A, 100B, 100C, may be shared with the dispatching server computing devices and/or other vehicles of the fleet. This information may be used to update the model or in conjunction with the lists of qualities to rank the regions. This information

could also be used to directly update the values for the lists of qualities *(Examiner notes; list of qualities can also be a plurality of permissible stopping location criteria)* or as some form of additional cost in the ranking. For example. if a region (such as a block of a street) typically has a high number of available pullover locations, but the block is full of parked vehicles, this information may be observed by one vehicle and shared with the server computing devices before another vehicle gets a request for a pick up or drop off within that region. As another example, if a vehicle blocks a driveway (that may have been considered rarely used) causes inconvenience to another road user, this information may be shared and used to avoid repeating the same behavior at the same driveway.").

Regarding claim 5, Modified Dyer teaches the method of claim 1 as discussed above. Dyer does not explicitly teach but Leary teaches, **further comprising, by the one or more processors after the one or more processors have received the departure confirmation: before the AV reaches the loading point;** (See Leary, **paragraph 0123**; "In some cases, the autonomous vehicle 102 can detect if the passenger 1010 has changed course or started to go on the wrong direction using GPS information relayed or shared by the client device 170 of the passenger 1010"); **processing data received from a perception system of the AV to determine whether the AV can stop and perform a loading operation at the loading point without obstructing another vehicle that is following the AV;** (See **Leary, paragraph 0072;** "The interface 320 can similarly include the map 302 with the drop-off options 304-310 identified for the user's trip. In this example, the drop-off location has been adjusted from drop-off option 304 to drop-off option 308. The drop-off location may be adjusted for one or more reasons such as, for example, changes in circumstances (e.g., traffic, weather, construction or other obstructions, user preference, street closures, nearby events, etc.). If the

user does not have further adjustments to make, the user can confirm the adjusted drop-off location through the interface element 312.");

otherwise navigating the AV to an intermediate stopping location and notifying the passenger electronic device that the loading operation will be performed at the intermediate stopping location; (See **Leary, paragraph 126**; "if an area along the path 1014 is blocked by construction or an event, the autonomous vehicle 102 can detect so using the one or more sensors 104. The autonomous vehicle 102 can then notify the passenger 1010 that the area along the path 1014 is blocked and provide instructions for avoiding the area and reaching the final destination 1016 from a deviated path.").

Both Dyer and Leary are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Leary autonomous vehicle rider drop-off to destination experience. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the autonomous vehicle rider drop-off to destination experience. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Dyer does not explicitly teach but Rasmusson teaches**, and only stopping the AV at the loading point to perform the loading operation if the loading operation will not obstruct the other vehicle;** (See **Rasmusson, paragraph 040**;" The space parameter may indicate a space on (or off) the road where vehicles may safely or legally stop. For example, a location along the shoulder of a road may be inside the space parameter, but a location in the middle of a highway

lane may be outside the space parameter. As another example, a location in a fire lane may be outside the space parameter, but a location within a bike lane may be inside the space parameter, since it may be illegal to stop in a fire lane but legal to stop in a bike lane. Any location of negative space inside the space parameter that is larger than the footprint of the autonomous vehicle (plus some buffer distance on all sides) may qualify as an available location for pick-up or drop-off. As an example and not by way of limitation, the autonomous-vehicle sensor data may indicate that there.").

Both Dyer and Rasmusson are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Rasmusson loading operation that does not obstruct the other vehicle. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the loading operation that does not obstruct the other vehicle. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 6, Modified Dyer teaches, the method of claim 5 as discussed above. Dyer further teaches, **, wherein determining whether the AV can stop and perform a loading operation at the loading point without obstructing another vehicle that is following the AV comprises: determining whether the loading point is in a location that will require the AV to at least partially block a lane in the other vehicle that is following the AV is currently traveling; and determining that the AV will obstruct the other vehicle if: the loading point is in a location that will require the AV to at least partially block the lane, and the loading**

operation will have a duration that exceeds a time before which the other vehicle can non-suddenly stop; (See **Dyer paragraph 0046;** "Vehicle inconvenience values may represent how much inconvenience the vehicle imposes on the other road users during the time the vehicle is pulled over by tracking the delta between the other road user's intent and the other road user's progress caused by the vehicle *(Examiner notes- tracking delta between the other road user's that would include vehicles behind the AV)*. This vehicle inconvenience value may be determined from prior attempts to pullover which may have caused problems to other road users *(Examiner notes – pullover causing problems to other users can include: partially block the lane, and the loading operation will have a duration that exceeds a time before which the other vehicle can non-suddenly stop)* .For instance, in the driveway example discussed above, if there are driveways that cause problems, these may be associated with a higher vehicle inconvenience)"; also See **Dyer paragraph 0073-0074;** "In some instances, the qualities of the highest ranked region may be compared by the computing devices 110 with qualities of a region in which the vehicle 100 is currently driving or pulled over in order to determine whether the vehicle should move to the highest ranked region. This may be helpful, for instance, if a vehicle is currently pulled over and waiting for a passenger, and the situation changes. As an example, if another vehicle approaches who may need to nudge around the vehicle, the vehicle's computing devices may do the comparison in order to determine whether it is beneficial to the vehicle, the passenger, and/or the another vehicle to move to another region.).

In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously

because of the inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation).").

Regarding claim 7, Modified Dyer teaches the method of claim 1 as discussed above. Dyer does not explicitly teach but Leary teaches**, further comprising, by the one or more processors after the one or more processors have received the departure confirmation and before the AV reaches the loading point;** (See Leary, **paragraph 0123**; "In some cases, the autonomous vehicle 102 can detect if the passenger 1010 has changed course or started to go on the wrong direction using GPS information relayed or shared by the client device 170 of the passenger 1010"); **processing data received from a perception system of the AV to: identify one or more objects located proximate to the loading point;** (See **Leary, paragraph 0035;**" The internal computing system 110 can also include a constraint service 114 to facilitate safe propulsion of the autonomous vehicle 102. The constraint service 116 includes instructions for activating a constraint based on a rule-based restriction upon operation of the autonomous vehicle 102. For example, the constraint may be a restriction on navigation that is activated in accordance with protocols configured to avoid occupying the same space as other objects, abide by traffic laws, circumvent avoidance areas, etc.");

otherwise navigate the AV to an intermediate stopping location and notifying the passenger electronic device that the loading operation will be performed at the intermediate stopping

location; (See **Leary, paragraph 126**; "if an area along the path 1014 is blocked by construction or an event, the autonomous vehicle 102 can detect so using the one or more sensors 104. The autonomous vehicle 102 can then notify the passenger 1010 that the area along the path 1014 is blocked and provide instructions for avoiding the area and reaching the final destination 1016 from a deviated path.").

Both Dyer and Leary are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Leary intermediate stopping location. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the intermediate stopping location. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Dyer does not explicitly teach but Rasmusson teaches**, measure a length of a stop zone that includes the loading point and that is free from obstruction use a current speed of the AV and the measured length of the stop zone to determine whether the AV can non-suddenly stop within the stop zone in compliance with the permissible stopping criteria;** (See **Rasmusson, paragraph 0040;** "The computing device may also identify and characterize locations at step 530. Identifying the locations may involve processing the autonomous-vehicle sensor data to determine the location of each object within a threshold distance of the autonomous vehicle (e.g., 100 feet) and then determining the locations and size of the negative spaces (e.g., the space between objects) inside a space parameter. The space parameter may indicate a space on (or off) the road where vehicles may safely or legally stop. For example, a

location along the shoulder of a road may be inside the space parameter, but a location in the middle of a highway lane may be outside the space parameter. As another example, a location in a fire lane may be outside the space parameter, but a location within a bike lane may be inside the space parameter, since it may be illegal to stop in a fire lane but legal to stop in a bike lane. Any location of negative space inside the space parameter that is larger than the footprint of the autonomous vehicle (plus some buffer distance on all sides) *(Examiner notes in order to determine if the space is larger than the vehicle the space has to be measured)* may qualify as an available location for pick-up or drop-off. As an example, and not by way of limitation, the autonomous-vehicle sensor data may indicate that there are three stationary objects within the space parameter along the side of a road. The distance between the first object and the second object may be a first negative space (e.g., empty space that is not occupied by any objects) that is 16 feet long. This may be a few inches shorter than the length of the autonomous vehicle. The distance between the second object and the third object may be a second negative space that is 20 feet long. This may be 3 feet and 8 inches longer than the length of the autonomous vehicle. The 3 feet and 8 inches may be greater than the buffer distance (e.g., 6 inches). Thus, the computing device may determine that the first negative space is not available for pick-up or drop-off because it is too small but that the second negative space is available. Thus, the computing device may identify the second negative space as an available location."); **and only stop at the loading point to perform the loading operation if the AV can non- suddenly stop within the stop zone in compliance with the permissible stopping criteria;** (See **Rasmusson, paragraph 0029**; "FIG. 2 illustrates an example driving environment of an example autonomous vehicle 140. In particular embodiments, a computing device associated with autonomous

vehicle 140 or dynamic transportation matching system 160 may receive autonomous-vehicle sensor data that represents an external environment within a threshold distance of autonomous vehicle 140. In particular embodiments, the computing device may be autonomous-vehicle UI device 148, may be navigation system 146, or may be any other suitable computing device associated with autonomous vehicle 140. The autonomous-vehicle sensor data may be collected via sensors arranged on the outside or the inside of autonomous vehicle 140. The autonomous-vehicle sensor data may enable autonomous vehicle 140 to identify objects in the surrounding external environment, such as vehicles 210. The autonomous-vehicle sensor data may further enable autonomous vehicle 140 to identify the road upon which it is driving, lanes in the road, or any other suitable object." *(Examiner notes - by scanning the area the vehicle determines if it non-suddenly stop within the stop zone)*).

Both Dyer and Rasmusson are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Rasmusson size of loading points that are free of obstruction. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the loading points that are free of obstruction. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 8, Modified Dyer teaches the method of claim 1 as discussed above. Dyer does not explicitly teach but Leary teaches**, by the one or more processors after the AV**

has stopped in the pickup area: receiving a nudge request from the passenger electronic device; (See **Leary, paragraph 0051**; "At block 204, the flow 200 can include determining (e.g., via the internal computing system 110 and/or the remote computing system 150) whether the user has adjusted the drop-off location for the trip. If the user has adjusted the drop-off location, at block 206, the flow 200 can include selecting (e.g., via the internal computing system 110 and/or the remote computing system 150) a new drop-off location based on the user's adjusted drop-off location. The flow 200 can then proceed to block 208, described below.");

and in response to determining that no objects would obstruct the AV when the AV would move to the position, update the loading point to move forward by the specified distance and move the AV to the updated loading point; (See **Leary, paragraph 0057**; "If the autonomous vehicle 102 does offer multiple pull-over options to the user, at block 216 the autonomous vehicle 102 can present (e.g., via the internal computing system 110 and/or the remote computing system 150) the various pull-over options to the user. In some examples, the autonomous vehicle 102 can present the various pull-over options to the user on display device 108 and/or the ridesharing application 172. The user can select a pull-over location from the pull-over options presented to the user.").

Both Dyer and Leary are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Leary determine whether any objects would obstruct the move forward from the loading point.

No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the determine whether any objects would obstruct the move forward from the loading point. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Dyer does not explicitly teach but Rasmussen teaches, **processing data received by a perception system of the AV to determine whether any objects would obstruct the AV when the AV would move to a position that is ahead of the loading point by a specified distance;** (See **Rasmusson, paragraph 0040**;" ...The computing device may also identify and characterize locations at step 530. Identifying the locations may involve processing the autonomous-vehicle sensor data to determine the location of each object within a threshold distance of the autonomous vehicle (e.g., 100 feet) and then determining the locations and size of the negative spaces (e.g., the space between objects) inside a space parameter. The space parameter may indicate a space on (or off) the road where vehicles may safely or legally stop...").

Both Dyer and Rasmusson are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Rasmusson determine whether any objects would obstruct the move forward from the loading point. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the determine whether any objects would obstruct the

move forward from the loading point. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 9, Modified Dyer teaches, the method of claim 1 as discussed above. Dyer further teaches**, further comprising, by the one or more processors when the one or more processors have not received the departure confirmation, determining the intermediate stopping location by: selecting a road segment that is a nearest useable road segment to the pickup location; identifying, from one or more permissible stopping points along the road segment, the permissible stopping point that is closest to the pickup location; and using the permissible stopping point that is closest to the pickup location as the intermediate stopping location;** (**See Dyer Paragraph 74;** "In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously because of the inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation)." ; **also see Dyer paragraph 0046;**"Other signals stored in the storage system 450 may only be available or determined by the server computing devices 410 and/or human operators after a pullover is actually attempted by one of the vehicles of the fleet. "; as well **see Dyer paragraph 0048;** "In order to build a model, the server computing devices 410 may access the signals of the storage system 450 described above. Using these signals or values determined from these signals, a model that identifies expected characteristics of regions may be built by the server computing devices 410. The model may be trained such that for a given geographic constraint (such as a specific location or a region) and time constraint (such as day of the year, calendar month, day of week, and/or time of day), the model may provide a list of expected

qualities for a region. In this regard, if a specific location is provided, the region may correspond to a region that includes the specific location. In this regard, the signals or values determined from these signals for a given region where the sensor data corresponding to the signals was collected may be used as training outputs for the model, and the specific location or given region and a time when the sensor data was collected may be used as training inputs to the model. In the case of a machine-learned model, the training may essentially tune parameter values for the model.").

Regarding claim 10, Dyer teaches, **a ride service system, comprising: an autonomous vehicle (AV) comprising a perception system, a motion planning system and a motion control system; a processor; and a memory containing programming instructions that are configured to cause the processor to determine a stopping location for the AV before picking up a passenger by;** (**See Dyer paragraph 0025**; " In one aspect the computing devices 110 may be part of an autonomous control system capable of communicating with various components of the vehicle in order to control the vehicle in an autonomous driving mode. For example, returning to FIG. 1, the computing devices 110 may be in communication with various systems of vehicle 100, such as deceleration system 160, acceleration system 162, steering system 164, routing system 166, planning system 168, positioning system 170, and perception system 172 in order to control the movement, speed, etc. of vehicle 100 in accordance with the instructions 134 of memory 130 in the autonomous driving mode."; also **See Dyer paragraph 0002;** "determine, by the one or more processors, whether to attempt to find a pullover location within the region to perform the pullover maneuver based on the list of qualities for the region; and maneuvering, by the one or more processors, the vehicle in the autonomous driving mode based on the determination whether to attempt to find a pullover location within the region." *(Examiner notes: pull over locations the same as stopping location)*); **receiving a ride service request that has been transmitted from a ride service application on a passenger electronic device, determining a pickup area for the ride service**

request, along with a loading point that is within a pickup area, causing the motion planning system to determine a route from a starting location of the AV to the pickup area, causing the motion control system to move the AV along the route toward the pickup area; (See Dyer paragraph 0042; "Although the client computing devices 420, 430, and 440 may each comprise a full-sized personal computing device, they may alternatively comprise mobile computing devices capable of wirelessly exchanging data with a server over a network such as the Internet. By way of example only, client computing device 420 may be a mobile phone or a device such as a wireless-enabled PDA, a tablet PC, a wearable computing device or system, or a netbook that is capable of obtaining information via the Internet or other networks. In another example, client computing device 430 may be a wearable computing system, shown as a wristwatch as shown in FIG. 3. As an example, the user may input information using a small keyboard, a keypad, microphone, using visual signals with a camera, or a touch screen.", also **See Dyer paragraph 0076;** " For instance, the server computing devices may use the model to identify a region for a given pickup or destination location for a trip (i.e. where a passenger wants to be picked up or dropped off) computing devices 110 may send a request to the server computing devices to identify a region given a current destination for the vehicle and/or pickup or drop off location for a passenger."); determining a route from a starting location of the AV to the pickup area; (**See Dyer paragraph 0027;** "Planning system 168 may be used by computing devices 110 in order to determine and follow a route generated by a routing system 166 to a location. For instance, the routing system 166 may use map information to determine a route from a current location of the vehicle to a drop off location. "). navigating the AV along the route toward the pickup area; (See **Dyer paragraph 0036;** "The computing devices 110 may control the vehicle in an autonomous driving mode by controlling various components. For instance, by way of example, the computing devices 110 may navigate the vehicle to a destination location completely autonomously using data from the detailed map information and planning system 168. ");

Dyer does not explicitly teach, but Leary teaches, **before the AV reaches the pickup area, determining whether the processor has received a departure confirmation from the passenger electronic device;** (See **Leary, paragraph 0078;** "In some cases, the tracking interface 420 can also include an interface element 428 that the user can select to end or cancel the trip. For example, the user can track the autonomous vehicle 102 through the tracking interface 420 and view tracking details. If the user decides to end or cancel the trip, the user can select the interface element 428, which would trigger a request or instruction to end or cancel the trip. The client device 170 can send a signal (e.g., via a network such as a cellular or WIFI network) to the autonomous vehicle 102 (e.g., to the internal computing system 110 and/or the remote computing system 150) indicating that the user has ended or canceled the trip."");

and if the one or more processors have received the departure confirmation, causing the motion control system to move the AV into the pickup area and stop at the loading point; (See **Leary, paragraph 0079**; "If the user does not end or cancel the trip, the autonomous vehicle 102 can continue and pick up the user. Once the user is picked up by the autonomous vehicle 102, the autonomous vehicle 102 can travel towards the drop-off location configured for the trip."; also see Leary, paragraph 0084; " FIG. 5B illustrates an example notification 520 sent to the user's client device 170 to inform the user that the autonomous vehicle 102 is searching for a pull-over location. As the autonomous vehicle 102 searches (and/or before) for a pull-over location, the autonomous vehicle 102 can send (e.g., via the internal computing system 110 and/or the remote computing system 150) the notification 520 to the client device 170 for presentation to the user at the client device 170. The notification 520 can indicate that the autonomous vehicle 102 is searching for a pull-over location. In some cases, the notification

520 can also provide other information such as, for example, an indication of one or more pull-over options identified by the autonomous vehicle 102, details about one or more pull-over options, an alert (e.g., a hazard alert, a weather alert, an update alert, etc.), a request to confirm a pull-over location, and/or any other information."); **otherwise causing the motion control system to either (a) move the AV to an intermediate stopping location before reaching the pickup area or to (b) cause the AV to pass through the pickup area;** (See **Leary, paragraph 0084;** " FIG. 5B illustrates an example notification 520 sent to the user's client device 170 to inform the user that the autonomous vehicle 102 is searching for a pull-over location. As the autonomous vehicle 102 searches (and/or before) for a pull-over location, the autonomous vehicle 102 can send (e.g., via the internal computing system 110 and/or the remote computing system 150) the notification 520 to the client device 170 for presentation to the user at the client device 170. The notification 520 can indicate that the autonomous vehicle 102 is searching for a pull-over location. In some cases, the notification 520 can also provide other information such as, for example, an indication of one or more pull-over options identified by the autonomous vehicle 102, details about one or more pull-over options, an alert (e.g., a hazard alert, a weather alert, an update alert, etc.), a request to confirm a pull-over location, and/or any other information."; also **see Leary paragraph 0082;** "In some cases, the map 502 can also depict other pull-over options 506B-N for the user. The other pull-over options 506B-N can provide alternatives that the user may select from. If the user prefers one of the other pull-over options 506B-N more than the pull-over location 506A, the user can select a particular pull-over option 506B-N from the map 502 to trigger the autonomous vehicle 102 to drop off the user at the other, selected pull-over option; and see **Leary paragraph 0084;** "FIG. 5B

illustrates an example notification 520 sent to the user's client device 170 to inform the user that the autonomous vehicle 102 is searching for a pull-over location. As the autonomous vehicle 102 searches (and/or before) for a pull-over location, the autonomous vehicle 102 can send (e.g., via the internal computing system 110 and/or the remote computing system 150) the notification 520 to the client device 170 for presentation to the user at the client device 170. The notification 520 can indicate that the autonomous vehicle 102 is searching for a pull-over location. In some cases, the notification 520 can also provide other information such as, for example, an indication of one or more pull-over options identified by the autonomous vehicle 102, details about one or more pull-over options, an alert (e.g., a hazard alert, a weather alert, an update alert, etc.), a request to confirm a pull-over location, and/or any other information." *(Examiner notes, vehicle searching for pull-over location, choosing pull-over location from multiple choices and sending the notification to the client device about chosen pull-over location indicates that the vehicle is stopping intermediate stopping location (one of the pull-over locations) or pass through the pickup area (first pull-over location that might not be available.)*).

Both Dyer and Leary are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Leary navigating AV along the route. No new functionality would arise from the combination and the
combination would improve usability of Dyer by adding the navigating AV along the route. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Dyer does not explicitly teach, but Rasmusson teaches, **wherein the departure confirmation comprises data confirming that coordinates of the passenger electronic device correspond to the loading point;** (See **Rasmusson, paragraph 0038;** "At step 520, the computing device determines whether the autonomous vehicle is within a threshold proximity of a current path of a current ride of the autonomous vehicle. The current path may include the ride origin or the ride destination. The ride origin may be the location where the autonomous vehicle picks up the requestor…In particular embodiments, the ride origin includes GPS coordinates of the client device 130 associated with the requestor, a request location entered into a requestor device…").

Both Dyer and Rasmusson are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer autonomous vehicle rider drop-off to destination experience with Rasmusson passenger electronic device coordinates that correspond to the loading point. No new functionality would arise from the combination and the combination would improve usability of Leary by adding the passenger electronic device coordinates that correspond to the loading point. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 11, Modified Dyer teaches the method of claim 10 as discussed above.

Dyer does not explicitly teach, but Leary teaches, **and use the loading point location rules to select the loading point from a set of candidates loading points;** (**See Leary, paragraph 0051;** "At block 204, the flow 200 can include determining (e.g., via the internal computing system 110 and/or the remote computing system 150) whether the user has adjusted the drop-off location for the trip. If the user has adjusted the drop-off location, at block 206, the flow 200 can include selecting (e.g., via the internal computing system 110 and/or the remote computing system 150) a new drop-off location based on the user's adjusted drop-off location. The flow 200 can then proceed to block 208, described below.").

Both Dyer and Leary are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Leary autonomous vehicle rider drop-off to destination experience. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the autonomous vehicle rider drop-off to destination experience. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Dyer does not explicitly teach, but Rasmusson teaches, **wherein the instructions to determine the loading point comprise instructions to: determine a passenger identifier that is associated with the passenger electronic device;** (See **Rasmusson, paragraph 0015;** " When dynamic transportation matching system 160 receives a ride request, it may also receive an identifier from a user device 130 associated with user 101. The identifier may identify user 101 to dynamic transportation matching system 160 for the purpose of accessing rider information

about user 101 in accordance with the rider's privacy settings."); **access a passenger profile server and query the passenger profile server to return a passenger profile that corresponds to the passenger identifier; access the returned passenger profile to extract one or more loading point location rules from the returned passenger profile; (See Rasmusson, paragraph 0015;** "The identifier may identify user 101 to dynamic transportation matching system 160 for the purpose of accessing rider information about user 101 in accordance with the rider's privacy settings. Rider information may be stored on one or more data stores associated by dynamic transportation matching system 160. In particular embodiments, rider information may include information about a particular rider or information about riders in the aggregate. Such information may include preferred pick-up or drop-off locations for a particular rider or for a threshold number of riders, one or more autonomous-vehicle settings of the rider or riders (such as preferred speed, preferred rates of acceleration/deceleration, preferred following distance when travelling at various speeds, etc.), music settings (e.g. preferred playlists, volume, etc.), frequent destinations for a particular rider, payment information, or any other suitable information.").

Both Dyer and Rasmusson are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Rasmusson confirmation, passenger identifier and ride instructions. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the confirmation, passenger identifier and ride instructions. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 12, Modified Dyer teaches, the ride service system of claim 11 as discussed above. Dyer further teaches, **wherein the loading point location rules comprise one or more rules to prioritize one or more of the following characteristics to determine the loading point: a lane segment that is in front of an entry of a building having a location that corresponds to a location of the passenger electronic device; a lane segment that is midway between two cross streets; or a lane segment that is as close as possible to the cross street that is next in the direction of traffic while remaining a threshold distance away from that cross street;** (See **Dyer paragraph 0073 – 0075**; "In some instances, the qualities of the highest ranked region may be compared by the computing devices 110 with qualities of a region in which the vehicle 100 is currently driving or pulled over in order to determine whether the vehicle should move to the highest ranked region. This may be helpful, for instance, if a vehicle is currently pulled over and waiting for a passenger, and the situation changes. As an example, if another vehicle approaches who may need to nudge around the vehicle, the vehicle's computing devices may do the comparison in order to determine whether it is beneficial to the vehicle, the passenger, and/or the another vehicle to move to another region.

In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously because of the inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation).

In some instances, real time information about the availability of pullover locations observed by a vehicle, such as any of vehicles 100, 100A, 100B, 100C, may be shared with the dispatching server computing devices and/or other vehicles of the fleet. This information may be used to update the model

or in conjunction with the lists of qualities to rank the regions. This information could also be used to directly update the values for the lists of qualities or as some form of additional cost in the ranking. For example. if a region (such as a block of a street) typically has a high number of available pullover locations, but the block is full of parked vehicles, this information may be observed by one vehicle and shared with the server computing devices before another vehicle gets a request for a pick up or drop off within that region. As another example, if a vehicle blocks a driveway (that may have been considered rarely used) causes inconvenience to another road user, this information may be shared and used to avoid repeating the same behavior at the same driveway.").

Regarding claim 13, Modified Dyer teaches, the ride service system of claim 10 as discussed above. **Dyer further teaches, further comprising additional programming instructions that are configured to cause the processor to determine the intermediate stopping location as a location that: is along the route between a current location of the AV and the loading point; and satisfies each of a plurality of permissible stopping location criteria;** (**See Dyer paragraph 0075;** "In some instances, real time information about the availability of pullover locations observed by a vehicle, such as any of vehicles 100, 100A, 100B, 100C, may be shared with the dispatching server computing devices and/or other vehicles of the fleet. This information may be used to update the model or in conjunction with the lists of qualities to rank the regions. This information could also be used to directly update the values for the lists of qualities *(Examiner notes; list of qualities can also be a plurality of permissible stopping location criteria)* or as some form of additional cost in the ranking. For example. if a region (such as a block of a street) typically has a high number of available pullover locations, but the block is full of parked vehicles, this information may be observed by one vehicle and shared with the server computing devices before another vehicle gets a request for a pick up or drop off within that region. As another example, if a vehicle blocks a driveway (that may have been considered rarely used) causes

inconvenience to another road user, this information may be shared and used to avoid repeating the same behavior at the same driveway.").

Regarding claim 14, Modified Dyer teaches, **the ride service system of claim 10, further comprising additional programming instructions that are configured to cause the processor to, upon receipt of the departure confirmation: before the AV reaches the loading point, process data received from a perception system of the AV to determine whether the AV can stop and perform a loading operation at the loading point without obstructing another vehicle that is following the AV;** (See **Dyer paragraph 0074;** "In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously because of the inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation)."; **and only stop the AV at the loading point to perform the loading operation if the loading operation will not obstruct the another other vehicle; see Dyer paragraph 0073;** " ... if another vehicle approaches who may need to nudge around the vehicle, the vehicle's computing devices may do the comparison in order to determine whether it is beneficial to the vehicle, the passenger, and/or the another vehicle to move to another region.").

Dyer does not explicitly teach, but Leary teaches, **otherwise cause the AV to navigate to an intermediate stopping location and notify the passenger electronic device that the loading operation will be performed at the intermediate stopping location;** (See **Leary, paragraph 126**; "if an area along the path 1014 is blocked by construction or an event, the autonomous vehicle 102 can detect so using the one or more sensors 104. The autonomous vehicle 102 can then notify the passenger 1010 that the area along the path 1014 is blocked and provide instructions for avoiding the area and reaching the final destination 1016 from a deviated path.").

Both Dyer and Leary are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Leary autonomous vehicle rider drop-off to destination experience. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the autonomous vehicle rider drop-off to destination experience. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 15, Modified Dyer teaches, the ride service system of claim 14 as discussed above. Dyer further teaches, **wherein the instructions to determine whether the AV can stop and perform a loading operation at the loading point without obstructing another vehicle that is following the AV comprise instructions to: determine whether the loading point is in a location that will require the AV to at least partially block a lane in the other vehicle that is following the AV is currently traveling; and determine that the AV will obstruct the other vehicle if: the loading point is in a location that will require the AV to at least partially block the lane, and the loading operation will**

have a duration that exceeds a time before which the other vehicle can non-suddenly stop; (See **Dyer paragraph 0046;** "Vehicle inconvenience values may represent how much inconvenience the vehicle imposes on the other road users during the time the vehicle is pulled over by tracking the delta between the other road user's intent and the other road user's progress caused by the vehicle *(Examiner notes- tracking delta between the other road user's that would include vehicles behind the AV)*. This vehicle inconvenience value may be determined from prior attempts to pullover which may have caused problems to other road users *(Examiner notes – pullover causing problems to other users can include: partially block the lane, and the loading operation will have a duration that exceeds a time before which the other vehicle can non-suddenly stop)* .For instance, in the driveway example discussed above, if there are driveways that cause problems, these may be associated with a higher vehicle inconvenience)"; also See **Dyer paragraph 0073-0074;** "In some instances, the qualities of the highest ranked region may be compared by the computing devices 110 with qualities of a region in which the vehicle 100 is currently driving or pulled over in order to determine whether the vehicle should move to the highest ranked region. This may be helpful, for instance, if a vehicle is currently pulled over and waiting for a passenger, and the situation changes. As an example, if another vehicle approaches who may need to nudge around the vehicle, the vehicle's computing devices may do the comparison in order to determine whether it is beneficial to the vehicle, the passenger, and/or the another vehicle to move to another region. In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously because of the inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation).").

Both Leary and Dyer are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Leary autonomous vehicle rider drop-off to destination experience with Dyer loading operation without obstructing another vehicle that is following the AV. No new functionality would arise from the combination and the combination would improve usability of Leary by adding the loading operation without obstructing another vehicle that is following the AV. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 16, Modified Dyer teaches the method of claim 10 as discussed above. Dyer does not explicitly teach but Leary teaches, **further comprising additional programming instructions that are configured to cause the processor to, after receiving the departure confirmation and before the AV reaches the loading point;** (See Leary, **paragraph 0123**; "In some cases, the autonomous vehicle 102 can detect if the passenger 1010 has changed course or started to go on the wrong direction using GPS information relayed or shared by the client device 170 of the passenger 1010"); **processing data received from the perception system of the AV to: identify one or more objects located proximate to the loading point;** (See **Leary, paragraph 0035;**" The internal computing system 110 can also include a constraint service 114 to facilitate safe propulsion of the autonomous vehicle 102. The constraint service 116 includes instructions for activating a constraint based on a rule-based restriction upon operation of the autonomous vehicle 102. For example, the constraint may be a restriction on navigation that is activated in accordance with protocols configured to avoid occupying the same space as other objects, abide by traffic laws, circumvent avoidance areas, etc."); **otherwise navigate the AV to an intermediate stopping location and notifying the passenger electronic device that the loading operation will be performed at the intermediate stopping**

location; (See **Leary, paragraph 126**; "if an area along the path 1014 is blocked by construction or an event, the autonomous vehicle 102 can detect so using the one or more sensors 104. The autonomous vehicle 102 can then notify the passenger 1010 that the area along the path 1014 is blocked and provide instructions for avoiding the area and reaching the final destination 1016 from a deviated path.").

Both Dyer and Leary are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Leary request confirmation, passenger identifier and ride instructions. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the request confirmation, passenger identifier and ride instructions. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Dyer does not explicitly teach but Rasmusson teaches, **measure a length of a stop zone that includes the loading point and that is free from obstruction; use a current speed of the AV and the measured length of the stop zone to determine whether the AV can non-suddenly stop within the stop zone while complying with permissible stopping criteria;** (See **Rasmusson, paragraph 0040;** "The computing device may also identify and characterize locations at step 530. Identifying the locations may involve processing the autonomous-vehicle sensor data to determine the location of each object within a threshold distance of the autonomous vehicle (e.g., 100 feet) and then determining the locations and size of the negative spaces (e.g., the space between objects) inside a space parameter. The space parameter may

indicate a space on (or off) the road where vehicles may safely or legally stop. For example, a location along the shoulder of a road may be inside the space parameter, but a location in the middle of a highway lane may be outside the space parameter. As another example, a location in a fire lane may be outside the space parameter, but a location within a bike lane may be inside the space parameter, since it may be illegal to stop in a fire lane but legal to stop in a bike lane. Any location of negative space inside the space parameter that is larger than the footprint of the autonomous vehicle (plus some buffer distance on all sides) may qualify as an available location for pick-up or drop-off. As an example and not by way of limitation, the autonomous-vehicle sensor data may indicate that there are three stationary objects within the space parameter along the side of a road. The distance between the first object and the second object may be a first negative space (e.g., empty space that is not occupied by any objects) that is 16 feet long. This may be a few inches shorter than the length of the autonomous vehicle. The distance between the second object and the third object may be a second negative space that is 20 feet long. This may be 3 feet and 8 inches longer than the length of the autonomous vehicle. The 3 feet and 8 inches may be greater than the buffer distance (e.g., 6 inches). Thus, the computing device may determine that the first negative space is not available for pick-up or drop-off because it is too small but that the second negative space is available. Thus, the computing device may identify the second negative space as an available location."); **and only stop at the loading point to perform the loading operation if the AV can non- suddenly stop within the stop zone while complying with the permissible stopping criteria;** (See **Rasmusson, paragraph 0029**; "FIG. 2 illustrates an example driving environment of an example autonomous vehicle 140. In particular embodiments, a computing device associated with autonomous

vehicle 140 or dynamic transportation matching system 160 may receive autonomous-vehicle sensor data that represents an external environment within a threshold distance of autonomous vehicle 140. In particular embodiments, the computing device may be autonomous-vehicle UI device 148, may be navigation system 146, or may be any other suitable computing device associated with autonomous vehicle 140. The autonomous-vehicle sensor data may be collected via sensors arranged on the outside or the inside of autonomous vehicle 140. The autonomous-vehicle sensor data may enable autonomous vehicle 140 to identify objects in the surrounding external environment, such as vehicles 210. The autonomous-vehicle sensor data may further enable autonomous vehicle 140 to identify the road upon which it is driving, lanes in the road, or any other suitable object." *(Examiner notes - by scanning the area the vehicle determines if it non-suddenly stop within the stop zone)*).

Both Dyer and Rasmusson are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Rasmusson size of loading points that are free of obstruction. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the loading points that are free of obstruction. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 17, Modified Dyer teaches the method of claim 10 as discussed above. Dyer does not explicitly teach but Leary teaches, **further comprising additional programming instructions**

that are configured to cause the processor to, after the AV has stopped in the pickup area and in response to receiving a nudge request from the passenger electronic device: (See **Leary, paragraph 0051**; "At block 204, the flow 200 can include determining (e.g., via the internal computing system 110 and/or the remote computing system 150) whether the user has adjusted the drop-off location for the trip. If the user has adjusted the drop-off location, at block 206, the flow 200 can include selecting (e.g., via the internal computing system 110 and/or the remote computing system 150) a new drop-off location based on the user's adjusted drop-off location. The flow 200 can then proceed to block 208, described below.");

and in response to determining that no objects would obstruct the AV when the AV would move to the position, update the loading point to move forward by the specified distance and move the AV to the updated loading point; (See **Leary, paragraph 0057**; "If the autonomous vehicle 102 does offer multiple pull-over options to the user, at block 216 the autonomous vehicle 102 can present (e.g., via the internal computing system 110 and/or the remote computing system 150) the various pull-over options to the user. In some examples, the autonomous vehicle 102 can present the various pull-over options to the user on display device 108 and/or the ridesharing application 172. The user can select a pull-over location from the pull-over options presented to the user.").

Both Dyer and Leary are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Leary autonomous vehicle rider drop-off to destination experience. No new functionality would arise from the combination and the combination would improve usability of Dyer Leary

autonomous vehicle rider drop-off to destination experience. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Dyer does not explicitly teach but Rasmussen teaches, **process data received by the perception system of the AV to determine whether any objects would obstruct the AV when the AV would move to a position that is ahead of the loading point by a specified distance;** (See **Rasmusson, paragraph 0040**;" ...The computing device may also identify and characterize locations at step 530. Identifying the locations may involve processing the autonomous-vehicle sensor data to determine the location of each object within a threshold distance of the autonomous vehicle (e.g., 100 feet) and then determining the locations and size of the negative spaces (e.g., the space between objects) inside a space parameter. The space parameter may indicate a space on (or off) the road where vehicles may safely or legally stop...").

Both Dyer and Rasmusson are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Rasmusson determine whether any objects would obstruct the move forward from the loading point. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the determine whether any objects would obstruct the move forward from the loading point. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 18, Modified Dyer teaches, the ride service system of claim 10 as discussed above. Dyer further teaches, **further comprising additional programming instructions that are configured to cause the one or more processors to, in response to not receiving the departure confirmation, determine the intermediate stopping location by: selecting a road segment that is a nearest useable road segment to the pickup location; identifying, from one or more permissible stopping points along the road segment, the permissible stop ping point that is closest to the pickup location; and using the permissible stopping point that is closest to the pickup location as the intermediate stopping location;** (See **Dyer Paragraph 0074;** "In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously because of the inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation)." ; also **see Dyer paragraph 0046;** "Other signals stored in the storage system 450 may only be available or determined by the server computing devices 410 and/or human operators after a pullover is actually attempted by one of the vehicles of the fleet. "; as well **see Dyer paragraph 0048;** "In order to build a model, the server computing devices 410 may access the signals of the storage system 450 described above. Using these signals or values determined from these signals, a model that identifies expected characteristics of regions may be built by the server computing devices 410. The model may be trained such that for a given geographic constraint (such as a specific location or a region) and time constraint (such as day of the year, calendar month, day of week, and/or time of day), the model may provide a list of expected qualities for a region. In this regard, if a specific location is provided, the region may correspond to a region that includes the specific location. In

this regard, the signals or values determined from these signals for a given region where the sensor data corresponding to the signals was collected may be used as training outputs for the model, and the specific location or given region and a time when the sensor data was collected may be used as training inputs to the model. In the case of a machine-learned model, the training may essentially tune parameter values for the model.").

Regarding claim 19, Dyer teaches, **a method of determining a stopping location for a vehicle before picking up a passenger, the method comprising: by one or more processors, at least one of which is a component of an autonomous vehicle (AV):** (See **Dyer paragraph 0002;** "One aspect of the disclosure provides a method of maneuvering a vehicle in an autonomous driving mode; identifying, by the one or more processors, a geographic constraint for the pullover maneuver … and maneuvering, by the one or more processors, the vehicle in the autonomous driving mode based on the determination whether to attempt to find a pullover location within the region." *(Examiner notes: pull over locations the same as stopping location)*); **receiving a ride service request that has been transmitted from a ride service application on a passenger electronic device; determining a pickup area for the ride service request, along with a loading point that is within a pickup area;** (See **Dyer paragraph 0042;** "…By way of example only, client computing device 420 may be a mobile phone or a device such as a wireless-enabled PDA, a tablet PC, a wearable computing device or system, or a netbook that is capable of obtaining information via the Internet or other networks. In another example, client computing device 430 may be a wearable computing system, shown as a wristwatch as shown in FIG. 3. As an example, the user may input information using a small keyboard, a keypad, microphone, using visual signals with a camera, or a touch screen.", also **See Dyer paragraph**

0076; " For instance, the server computing devices may use the model to identify a region for a given pickup or destination location for a trip (i.e. where a passenger wants to be picked up or dropped off) computing devices 110 may send a request to the server computing devices to identify a region given a current destination for the vehicle and/or pickup or drop off location for a passenger."); **determining a route from a starting location of the AV to the pickup area; navigating the AV along the route toward the pickup area;** (See **Dyer, paragraph 0027**; "Planning system 168 may be used by computing devices 110 in order to determine and follow a route generated by a routing system 166 to a location. For instance, the routing system 166 may use map information to determine a route from a current location of the vehicle to a drop off location. The planning system 168 may periodically generate trajectories, or short-term plans for, controlling the vehicle for some period of time into the future, in order to follow the route to the destination."; also **see Dyer Figure 10** block 1050);

determining an intermediate stopping location; (See **Leary, paragraph 0058**; "At block 218, the autonomous vehicle 102 can determine (e.g., using the internal computing system 110 and/or the remote computing system 150) whether the user has adjusted the pull-over location for the trip (e.g., based on a selected pull-over option from the various pull-over options presented to the user). If the user has not adjusted the pull-over location, the flow 200 can proceed to block 222, where the autonomous vehicle 102 selects the pull-over location where the user can exit the vehicle. Alternatively, at block 220 the autonomous vehicle 102 can adjust the pull-over location based on a determination that the user has requested to adjust the pull-over location. The flow 200 can then proceed to block 222, where the adjusted pull-over location is selected as the drop-off point for the user."); **as a location that (a) is along the route between a current**

location of the AV and the loading point, and (b) satisfies each of a plurality of permissible stopping location criteria; (See **Leary, paragraph 0084;** " FIG. 5B illustrates an example notification 520 sent to the user's client device 170 to inform the user that the autonomous vehicle 102 is searching for a pull-over location. As the autonomous vehicle 102 searches (and/or before) for a pull-over location, the autonomous vehicle 102 can send (e.g., via the internal computing system 110 and/or the remote computing system 150) the notification 520 to the client device 170 for presentation to the user at the client device 170. The notification 520 can indicate that the autonomous vehicle 102 is searching for a pull-over location. In some cases, the notification 520 can also provide other information such as, for example, an indication of one or more pull-over options identified by the autonomous vehicle 102, details about one or more pull-over options, an alert (e.g., a hazard alert, a weather alert, an update alert, etc.), a request to confirm a pull-over location, and/or any other information."; also **see Leary paragraph 0082;** "In some cases, the map 502 can also depict other pull-over options 506B-N for the user. The other pull-over options 506B-N can provide alternatives that the user may select from. If the user prefers one of the other pull-over options 506B-N more than the pull-over location 506A, the user can select a particular pull-over option 506B-N from the map 502 to trigger the autonomous vehicle 102 to drop off the user at the other, selected pull-over option; and see **Leary paragraph 0084;** "FIG. 5B illustrates an example notification 520 sent to the user's client device 170 to inform the user that the autonomous vehicle 102 is searching for a pull-over location. As the autonomous vehicle 102 searches (and/or before) for a pull-over location, the autonomous vehicle 102 can send (e.g., via the internal computing system 110 and/or the remote computing system 150) the notification 520 to the client device 170 for

presentation to the user at the client device 170. The notification 520 can indicate that the autonomous vehicle 102 is searching for a pull-over location. In some cases, the notification 520 can also provide other information such as, for example, an indication of one or more pull-over options identified by the autonomous vehicle 102, details about one or more pull-over options, an alert (e.g., a hazard alert, a weather alert, an update alert, etc.), a request to confirm a pull-over location, and/or any other information." *(Examiner notes, vehicle searching for pull-over location, choosing pull-over location from multiple choices and sending the notification to the client device about chosen pull-over location indicates that the vehicle is stopping intermediate stopping location (one of the pull-over locations) or pass through the pickup area (first pull-over location that might not be available.)*);

and in response to determining that the system has received the departure confirmation: causing the AV to navigate toward the pickup area; (See **Leary, paragraph 0079**; "If the user does not end or cancel the trip, the autonomous vehicle 102 can continue and pick up the user. Once the user is picked up by the autonomous vehicle 102, the autonomous vehicle 102 can travel towards the drop-off location configured for the trip."; also see Leary, paragraph 0084; " FIG. 5B illustrates an example notification 520 sent to the user's client device 170 to inform the user that the autonomous vehicle 102 is searching for a pull-over location. As the autonomous vehicle 102 searches (and/or before) for a pull-over location, the autonomous vehicle 102 can send (e.g., via the internal computing system 110 and/or the remote computing system 150) the notification 520 to the client device 170 for presentation to the user at the client device 170. The notification 520 can indicate that the autonomous vehicle 102 is searching for a pull-over location. In some cases, the notification 520 can also provide other information such as,

for example, an indication of one or more pull-over options identified by the autonomous vehicle 102, details about one or more pull-over options, an alert (e.g., a hazard alert, a weather alert, an update alert, etc.), a request to confirm a pull-over location, and/or any other information."); **identifying one or more objects located proximate to the loading point;** (See **Leary, paragraph 0035;**" ... The constraint service 116 includes instructions for activating a constraint based on a rule-based restriction upon operation of the autonomous vehicle 102. For example, the constraint may be a restriction on navigation that is activated in accordance with protocols configured to avoid occupying the same space as other objects, abide by traffic laws, circumvent avoidance areas, etc.").

Both Dyer and Leary are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Leary navigating AV along the route. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the navigating AV along the route. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Dyer does not explicitly teach but Rasmusson teaches, **before the AV reaches the pickup area, determining whether the one or more processors have received a departure confirmation from the passenger electronic device;** (See **Rasmusson, paragraph 0015**; "In particular embodiments, dynamic transportation matching system 160 may facilitate transportation for one or more users 101. Dynamic transportation matching system 160 may

receive any number of ride requests from any number of users 101. When dynamic transportation matching system 160 receives a ride request, it may also receive an identifier from a user device 130 associated with user 101. The identifier may identify user 101 to dynamic transportation matching system 160 for the purpose of accessing rider information about user 101 in accordance with the rider's privacy settings."); **wherein the departure confirmation comprises data confirming that coordinates of the passenger electronic device correspond to the loading point;** (See **Rasmusson, paragraph 0038;** "At step 520, the computing device determines whether the autonomous vehicle is within a threshold proximity of a current path of a current ride of the autonomous vehicle. The current path may include the ride origin or the ride destination. The ride origin may be the location where the autonomous vehicle picks up the requestor…In particular embodiments, the ride origin includes GPS coordinates of the client device 130 associated with the requestor, a request location entered into a requestor device…");

measuring a length of a stop zone that includes the loading point and that is free from obstruction, using a current speed of the AV and the measured length of the stop zone to determine whether the AV can non-suddenly stop within the stop zone while satisfying each of the plurality of permissible stopping location criteria, and stopping at the loading point to perform the loading operation if the system determines that the AV can non-suddenly stop within the stop zone while satisfying each of the plurality of permissible stopping location criteria; (See **Rasmusson, paragraph 0040;** "The computing device may also identify and characterize locations at step 530. Identifying the locations may involve processing the autonomous-vehicle sensor data to determine the location of each object within a threshold

distance of the autonomous vehicle (e.g., 100 feet) and then determining the locations and size of the negative spaces (e.g., the space between objects) inside a space parameter. The space parameter may indicate a space on (or off) the road where vehicles may safely or legally stop. For example, a location along the shoulder of a road may be inside the space parameter, but a location in the middle of a highway lane may be outside the space parameter. As another example, a location in a fire lane may be outside the space parameter, but a location within a bike lane may be inside the space parameter, since it may be illegal to stop in a fire lane but legal to stop in a bike lane. Any location of negative space inside the space parameter that is larger than the footprint of the autonomous vehicle (plus some buffer distance on all sides) may qualify as an available location for pick-up or drop-off. As an example and not by way of limitation, the autonomous-vehicle sensor data may indicate that there are three stationary objects within the space parameter along the side of a road. The distance between the first object and the second object may be a first negative space (e.g., empty space that is not occupied by any objects) that is 16 feet long. This may be a few inches shorter than the length of the autonomous vehicle. The distance between the second object and the third object may be a second negative space that is 20 feet long. This may be 3 feet and 8 inches longer than the length of the autonomous vehicle. The 3 feet and 8 inches may be greater than the buffer distance (e.g., 6 inches). Thus, the computing device may determine that the first negative space is not available for pick-up or drop-off because it is too small but that the second negative space is available. Thus, the computing device may identify the second negative space as an available location.");**otherwise navigating the AV to the intermediate stopping location and notifying the passenger electronic device that the loading operation will be performed at**

the intermediate stopping location; (See **Leary, paragraph 126**; "if an area along the path 1014 is blocked by construction or an event, the autonomous vehicle 102 can detect so using the one or more sensors 104. The autonomous vehicle 102 can then notify the passenger 1010 that the area along the path 1014 is blocked and provide instructions for avoiding the area and reaching the final destination 1016 from a deviated path.").

Both Dyer and Rasmusson are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer autonomous vehicle rider drop-off to destination experience with Rasmusson passenger electronic device coordinates that correspond to the loading point. No new functionality would arise from the combination and the combination would improve usability of Leary by adding the passenger electronic device coordinates that correspond to the loading point. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 20, Modified Dyer teaches, the method of claim 19 as discussed above. Dyer further teaches, **further comprising, by the one or more processors after determining that the system has received the departure confirmation: before the AV reaches the loading point, processing data received from a perception system of the AV to determine whether the AV can stop and perform a loading operation at the loading point without obstructing another vehicle that is following the AV; and only stopping the AV at the loading point to perform the loading operation if the loading operation will not obstruct the another vehicle, otherwise navigating the AV to the intermediate stopping location and notifying the passenger electronic device that the loading operation will be**

performed at the intermediate stopping location; (See **Dyer paragraph 0074;** "In some instances, the computing devices 100 may possibly identify a region to travel to which may be less convenient to the passenger, but will now have a higher ranking than previously because of the inconvenience to the another vehicle. In such situations, if the vehicle is going to pick up a passenger, a notification may be sent to the passenger's client computing device identifying the highest ranked region and indicating that the vehicle is currently going to that region to drop off the passenger or that the vehicle is able to go to that region to pick up the passenger (and requesting confirmation)."; also **see Dyer paragraph 0073;** " In some instances, the qualities of the highest ranked region may be compared by the computing devices 110 with qualities of a region in which the vehicle 100 is currently driving or pulled over in order to determine whether the vehicle should move to the highest ranked region. This may be helpful, for instance, if a vehicle is currently pulled over and waiting for a passenger, and the situation changes. As an example, if another vehicle approaches who may need to nudge around the vehicle, the vehicle's computing devices may do the comparison in order to determine whether it is beneficial to the vehicle, the passenger, and/or another vehicle to move to another region.").

Regarding claim 21, Modified Leary teaches the method of claim 1 as discussed above. Leary does not explicitly teach but Rasmusson teaches, **wherein the data confirming that coordinates of the passenger electronic device correspond to the loading point comprise data confirming that the passenger electronic device is at or within a threshold distance from the loading point.;** (See **Rasmusson, paragraph 0038;** "At step 520, the computing device determines whether the autonomous vehicle is within a threshold proximity of a current path of a current ride of the autonomous vehicle. The current path may include the ride origin or the ride destination. The ride origin may be the location where the autonomous vehicle picks up

the requestor…In particular embodiments, the ride origin includes GPS coordinates of the client device 130 associated with the requestor, a request location entered into a requestor device…").

Both Dyer and Rasmusson are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Rasmusson passenger electronic device coordinates that correspond to the loading point. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the passenger electronic device coordinates that correspond to the loading point. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Regarding claim 22, Modified Leary teaches the method of claim 10 as discussed above. Leary does not explicitly teach but Rasmusson teaches, **wherein the data confirming that coordinates of the passenger electronic device correspond to the loading point comprise data confirming that the passenger electronic device is at or within a threshold distance from the loading point;** (See **Rasmusson, paragraph 0038;** "At step 520, the computing device determines whether the autonomous vehicle is within a threshold proximity of a current path of a current ride of the autonomous vehicle. The current path may include the ride origin or the ride destination. The ride origin may be the location where the autonomous vehicle picks up the requestor…In particular embodiments, the ride origin includes GPS coordinates of the client

device 130 associated with the requestor, a request location entered into a requestor device…").

Both Dyer and Rasmusson are in the same field of endeavor of autonomous vehicles control and navigation. It would have been obvious for one ordinary skilled in the art before the effective filing date of present invention to modify Dyer identification of pullover region with Rasmusson passenger electronic device coordinates that correspond to the loading point. No new functionality would arise from the combination and the combination would improve usability of Dyer by adding the passenger electronic device coordinates that correspond to the loading point. Further, finding that one of ordinary skill in the art would have recognized that the results of the combination were predictable.

Conclusion

THIS ACTION IS MADE FINAL. Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LIDIA KWIATKOWSKA whose telephone number is (571)272-5161. The examiner can normally be reached Monday-Friday 8:00-5:00.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Anne Antonucci can be reached on 3134466519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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